Crash insurance and the G10 carry premium: theory notes

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These notes prove every result listed in `theory/README.md` except R9 (inference), which is cited there with its conditions and is not proved here. Results that are standard are marked as such and attributed, and where a published proof follows the same route as the one written here, it is cited. The parts I claim as my own are stated at the start of each section. Throughout, Φ and φ are the standard normal distribution and density functions, $s = \sigma\sqrt{\tau}$, and for forward F and strike K

$$d_{\pm}(K) = \frac{\ln(F/K) \pm s^2/2}{s}.$$

The identity $K\varphi(d_-) = F\varphi(d_+)$ is used repeatedly.

R1. Domestic and foreign risk-neutral measures

This is the change of numeraire of Geman, El Karoui and Rochet (1995); the statement and proof are included for completeness. Girsanov's theorem and Itô's formula are standard results, stated for example in Shreve (2004).

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{Q}^d)$ satisfy the usual conditions, let B^d, B^f be strictly positive adapted money-market accounts with $B^d_0 = B^f_0 = 1$, and let $S > 0$ be adapted and càdlàg. Assume $Z_t = S_t B^f_t/(S_0 B^d_t)$ is a true $\mathbb{Q}^d$-martingale and define $d\mathbb{Q}^f/d\mathbb{Q}^d = Z_T$ on $\mathcal{F}_T$.

Proposition 1. *(a) $\mathbb{Q}^f$ is a probability measure equivalent to $\mathbb{Q}^d$. (b) For $\mathcal{F}_T$-measurable $X \geq 0$ or $X \in L^1(\mathbb{Q}^f)$, $\mathbb{E}^{\mathbb{Q}^f}[X \mid \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}^d}[Z_T X \mid \mathcal{F}_t]/Z_t$. (c) An adapted process Y with $Y_t \in L^1(\mathbb{Q}^f)$ is a $\mathbb{Q}^f$-martingale if and only if ZY is a $\mathbb{Q}^d$-martingale; in particular a foreign price V^f satisfies: V^f/B^f is a $\mathbb{Q}^f$-martingale if and only if $SV^f/B^d = S_0 Z\,(V^f/B^f)$ is a $\mathbb{Q}^d$-martingale. (d) If $dS_t/S_t = (r_d - r_f)\,dt + \sigma\,dW_t$ under $\mathbb{Q}^d$ with constant coefficients, then $W^f_t = W_t - \sigma t$ is a $\mathbb{Q}^f$-Brownian motion, $d(1/S_t) = (1/S_t)[(r_f - r_d)\,dt - \sigma\,dW^f_t]$, and $\mathbb{E}^{\mathbb{Q}^d} S_T = F$, $\mathbb{E}^{\mathbb{Q}^f}(1/S_T) = 1/F$ with $F = S_0 e^{(r_d - r_f)T}$.*

Proof. (a) $Z_T > 0$ and $\mathbb{E}^{\mathbb{Q}^d} Z_T = Z_0 = 1$, so $\mathbb{Q}^f$ is a probability measure; positivity of Z_T gives equivalence. (b) For $\mathcal{F}_t$-measurable $Y \geq 0$, the tower property and the martingale property give $\mathbb{E}^{\mathbb{Q}^f} Y = \mathbb{E}^{\mathbb{Q}^d}[Z_T Y] = \mathbb{E}^{\mathbb{Q}^d}[Z_t Y]$. For $A \in \mathcal{F}_t$,

$$\mathbb{E}^{\mathbb{Q}^f}[X\mathbf{1}_A] = \mathbb{E}^{\mathbb{Q}^d}\big[\mathbb{E}^{\mathbb{Q}^d}[Z_T X \mid \mathcal{F}_t]\mathbf{1}_A\big] = \mathbb{E}^{\mathbb{Q}^f}\Big[\frac{\mathbb{E}^{\mathbb{Q}^d}[Z_T X \mid \mathcal{F}_t]}{Z_t}\mathbf{1}_A\Big],$$

which characterises the conditional expectation. (c) Apply (b) on $[s,t]$: $\mathbb{E}^{\mathbb{Q}^f}[Y_t \mid \mathcal{F}_s] = \mathbb{E}^{\mathbb{Q}^d}[Z_t Y_t \mid \mathcal{F}_s]/Z_s$, so Y is a $\mathbb{Q}^f$-martingale exactly when $\mathbb{E}^{\mathbb{Q}^d}[Z_t Y_t \mid \mathcal{F}_s] = Z_s Y_s$. (d) Here $Z_t = \exp(\sigma W_t - \sigma^2 t/2)$, so Girsanov's theorem gives the first claim, and Itô's formula for $1/S$ gives the dynamics. The expectations follow from the lognormal law under each measure. □

Remark 2. Jensen's inequality gives $\mathbb{E}^{\mathbb{Q}^d}(1/S_T) > 1/\mathbb{E}^{\mathbb{Q}^d} S_T = 1/F$. There is no contradiction with (d): the foreign investor's forward price is an expectation under $\mathbb{Q}^f$, not $\mathbb{Q}^d$ (the paradox of Siegel, 1972). If Z is only a strict local martingale, then $\mathbb{E}Z_T < 1$ and Z_T does not define an

equivalent probability measure. Carr, Fisher and Ruf (2014) construct the foreign measure in that case as a Föllmer measure that is not equivalent to $\mathbb{Q}^d$, and restore the domestic–foreign symmetry with a modified pricing operator.

R2. Arbitrage-free call prices

The link between the second strike derivative of call prices and state prices is due to Breeden and Litzenberger (1978). The characterisation below is a known result: Föllmer and Schied (2004, Lemma 7.23) show that call prices consistent with a linear, positive pricing rule are the call prices of a unique probability measure with the forward as its mean, by the same argument through the right derivative of the call-price function and Fubini's theorem. I state it in the form used by the arbitrage checks and write out the proof for completeness.

Proposition 3. *Let $c : [0,\infty) \to \mathbb{R}$ and $F > 0$. There is a random variable $X \geq 0$ with $\mathbb{E}X = F$ and $c(K) = \mathbb{E}(X-K)^+$ for all $K \geq 0$ if and only if (i) c is convex, (ii) $c(0) = F$, (iii) $c'_+(0) \geq -1$ and (iv) $c(K) \to 0$ as $K \to \infty$. The law of X is then unique, with distribution function $G = 1 + c'_+$ on $[0,\infty)$ and $\mathbb{P}(X = 0) = 1 + c'_+(0)$.*

Proof. Necessity. Convexity of c follows from convexity of $K \mapsto (x-K)^+$ and linearity of expectation, and $c(0) = \mathbb{E}X = F$. Since $0 \leq (X-K)^+ \leq X \in L^1$ and $(X-K)^+ \to 0$ pointwise, dominated convergence gives (iv). For $h > 0$, the difference quotient $[(X-K-h)^+ - (X-K)^+]/h$ is bounded by 1 in absolute value and converges to $-\mathbf{1}\{X > K\}$ as $h \downarrow 0$, so $c'_+(K) = -\mathbb{P}(X > K)$. In particular $c'_+(0) = -\mathbb{P}(X > 0) \geq -1$.

Sufficiency. On $(0,\infty)$ the convex function c is continuous, its right derivative c'_+ is finite, nondecreasing and right-continuous, and $c(b) - c(a) = \int_a^b c'_+(u)\,du$ for $0 < a < b$. At the end point 0 convexity alone gives less: c can jump down at 0, with $c(0) > \lim_{K\downarrow 0} c(K)$, and then $c'_+(0) = -\infty$. Condition (iii) excludes this jump. The chord slope $(c(h) - c(0))/h$ is nondecreasing in $h > 0$ with limit $c'_+(0) \geq -1$, so $-h \leq c(h) - c(0) \leq h\,(c(1) - c(0))$ for $0 < h \leq 1$. Hence c is continuous at 0 and Lipschitz on every $[0,b]$, so it is absolutely continuous there and the identity $c(b) - c(a) = \int_a^b c'_+(u)\,du$ also holds with $a = 0$; and c'_+ is right-continuous at 0, because $c'_+(0) \leq c'_+(K) \leq (c(K') - c(K))/(K' - K)$ for $0 < K < K'$, and letting $K \downarrow 0$, by continuity at 0, and then $K' \downarrow 0$ gives $c'_+(K) \to c'_+(0)$. If $c'_+(K_0) > 0$ for some K_0, then $c(K) \geq c(K_0) + c'_+(K_0)(K - K_0) \to \infty$, contradicting (iv); so $c'_+ \leq 0$, c is nonincreasing and, by (iv), $c \geq 0$. Let $L = \lim_{K\to\infty} c'_+(K) \leq 0$. Since c'_+ is nondecreasing, $c'_+ \leq L$ everywhere, so $c(K) \leq F + LK$; $L < 0$ would force $c < 0$ for large K. Hence $L = 0$.

Define $G(K) = 0$ for $K < 0$ and $G(K) = 1 + c'_+(K)$ for $K \geq 0$. Then G is nondecreasing and right-continuous, $G(0) = 1 + c'_+(0) \geq 0$ by (iii), and $G(K) \to 1$. It is therefore the distribution function of a random variable $X \geq 0$ (Lebesgue–Stieltjes construction). By Tonelli's theorem,

$$\begin{aligned}\mathbb{E}(X-K)^+ &= \int_K^\infty \mathbb{P}(X > u)\,du = \int_K^\infty \big(1 - G(u)\big)du \\ &= -\int_K^\infty c'_+(u)\,du = c(K) - \lim_{b\to\infty} c(b) = c(K),\end{aligned}$$

and at $K = 0$, where the last step uses the identity on $[0,b]$ and hence the continuity of c at 0 that (iii) provides, $\mathbb{E}X = c(0) = F$. Uniqueness holds because c determines c'_+ and hence G. □

Remark 4. Extending c by $c(K) = F - K$ for $K < 0$, the second distributional derivative of c on $\mathbb{R}$ is the law of X; the jump of c' at 0 carries the atom $\mathbb{P}(X = 0)$. On a finite strike grid the conditions imply slopes in $[-1, 0]$ that are nondecreasing, which is the discrete test used in `src/qef/fx/arbitrage.py`; for finitely many strikes at one maturity, Davis and Hobson (2007, Theorem 3.1) give necessary and sufficient conditions of this form, with the zero-strike call included and prices strictly decreasing while positive. The implementation also applies

the density condition on implied total variance of Gatheral and Jacquier (2014, eq. (2.1) and Lemma 2.2).

R3. Spanning

The representation of a payoff by bonds, forwards and out-of-the-money options is the spanning formula of Carr and Madan (2001; first published 1998). The version below states explicit sufficient hypotheses, including the integrability condition under which expectations can be taken term by term. The proof is the standard Taylor expansion with integral remainder, written out under those hypotheses. In R3 and R4, X is the terminal exchange rate and $\mathbb{E}$ and $\mathbb{Q}$ refer to the same pricing measure.

Proposition 5. *Let $f : (0,\infty) \to \mathbb{R}$ have a locally absolutely continuous derivative f', so that f'' exists almost everywhere and is locally integrable. Let $X > 0$ almost surely with $\mathbb{E}X = F < \infty$, and write $P(K) = \mathbb{E}(K - X)^+$ and $C(K) = \mathbb{E}(X - K)^+$. If*

$$I = \int_0^F |f''(K)|\, P(K)\, dK + \int_F^\infty |f''(K)|\, C(K)\, dK < \infty,$$

then $f(X) \in L^1$ and

$$\mathbb{E}f(X) = f(F) + \int_0^F f''(K)\, P(K)\, dK + \int_F^\infty f''(K)\, C(K)\, dK.$$

Proof. Fix $x > 0$. Absolute continuity of f and f' on the interval between x and F gives $f(x) - f(F) = \int_F^x f'(u)\, du$ and $f'(u) = f'(F) + \int_F^u f''(K)\, dK$. Substituting and applying Fubini's theorem on the bounded interval (where f'' is integrable),

$$f(x) = f(F) + f'(F)(x - F) + \int_F^x f''(K)(x - K)\, dK.$$

For $x > F$ the last integral equals $\int_F^\infty f''(K)(x - K)^+ dK$; for $x < F$ it equals $\int_x^F f''(K)(K - x)\, dK = \int_0^F f''(K)(K - x)^+ dK$. Hence, for every $x > 0$,

$$f(x) = f(F) + f'(F)(x - F) + \int_0^F f''(K)(K - x)^+ dK + \int_F^\infty f''(K)(x - K)^+ dK.$$

The map $(K, \omega) \mapsto f''(K)(K - X(\omega))^+$ is jointly measurable. By Tonelli's theorem, $\mathbb{E}\int_0^F |f''(K)|(K - X)^+ dK = \int_0^F |f''(K)| P(K)\, dK$, and similarly for the call term, so both are at most $I < \infty$. Each term on the right is therefore integrable, hence so is $f(X)$, and Fubini's theorem allows expectation and integral to be exchanged. Since $\mathbb{E}(X - F) = 0$, the formula follows. □

Remark 6. For $f(x) = \ln(x/F)$, $f''(K) = -K^{-2}$, which gives the log contract $\mathbb{E}\ln(X/F) = -\int_0^F P(K)K^{-2}dK - \int_F^\infty C(K)K^{-2}dK$. For $f(x) = (\ln(x/F))^2$, $f''(K) = 2(1 + \ln(F/K))/K^2$, which changes sign at $K = eF$; the absolute-value condition in the proposition covers such weights. The moment contracts of Bakshi, Kapadia and Madan (2003, Theorem 1) are of this type; they expand around the spot price with a constant interest rate, whereas the contracts here are centred at the forward F. Quotes observe P and C only between the 10Δ strikes, so the tails of these integrals must be bounded (R4).

R4. Bounds on the unobserved tails

Jiang and Tian (2005, Proposition 2) bound the truncation error of model-free implied variance beyond the quoted strike range. The bounds below serve the same purpose under a different assumption, a power-type monotonicity of the distribution function in each tail, which makes them computable from the last quoted prices and extends them to the weights of the higher moment contracts. Lee (2004a) relates the number of finite moments of the underlying to the smallest slope of a linear bound, in log-moneyness, on total implied variance at extreme strikes in each wing, the asymptotic counterpart of the tail exponents used here. The lemma is elementary, an instance of the monotone form of l'Hôpital's rule, and with an exact power-law tail its bounds hold with equality: in the upper tail this is the relative pricing rule of Taleb, Yarckin, Mann, Delic and Spitznagel (2023, Section II-A, eq. (3)), and in the lower tail the power-law put extrapolation of Benaim, Dodgson and Kainth (2008). The hypotheses are tail-restricted forms of the star order (see R10 below). What I claim is the inequality under the weaker monotone-ratio hypotheses, the proposition that applies it to general and sign-changing weights and the resulting intervals; I have not found them in this form in the sources I consulted.

Let $G(u) = \mathbb{Q}(X \le u)$ and $\bar{G}(u) = \mathbb{Q}(X > u)$. By Tonelli's theorem, $P(K) = \int_0^K G(u)\,du$ and $C(K) = \int_K^\infty \bar{G}(u)\,du$. Let $K_{\min} < F < K_{\max}$ be the lowest and highest strikes at which prices are observed.

Lemma 7. *(a) If $u \mapsto G(u)/u^\gamma$ is nondecreasing on $(0, K_{\min}]$ for some $\gamma > 0$, then $P(K) \le P(K_{\min})(K/K_{\min})^{\gamma+1}$ for $0 < K \le K_{\min}$. (b) If $u \mapsto u^\eta \bar{G}(u)$ is nonincreasing on $[K_{\max}, \infty)$ for some $\eta > 1$, then $C(K) \le C(K_{\max})(K/K_{\max})^{1-\eta}$ for $K \ge K_{\max}$.*

Proof. (a) With $\lambda = K/K_{\min} \in (0, 1]$ and the substitution $u = \lambda v$, $P(K) = \lambda \int_0^{K_{\min}} G(\lambda v)\,dv$. Since $\lambda v \le v$ and $G(w)/w^\gamma$ is nondecreasing, $G(\lambda v) \le \lambda^\gamma G(v)$. Hence $P(K) \le \lambda^{\gamma+1} \int_0^{K_{\min}} G(v)\,dv = \lambda^{\gamma+1} P(K_{\min})$. (b) With $\lambda = K/K_{\max} \ge 1$ and $u = \lambda v$, $C(K) = \lambda \int_{K_{\max}}^\infty \bar{G}(\lambda v)\,dv$, and $\bar{G}(\lambda v) \le \lambda^{-\eta} \bar{G}(v)$ because $w^\eta \bar{G}(w)$ is nonincreasing. Hence $C(K) \le \lambda^{1-\eta} C(K_{\max})$. □

Proposition 8. *Under the hypotheses of the lemma, for any measurable $w \ge 0$ on $(0, K_{\min}]$,*

$$\int_0^{K_{\min}} w(K) P(K)\,dK \le \frac{P(K_{\min})}{K_{\min}^{\gamma+1}} \int_0^{K_{\min}} w(K) K^{\gamma+1} dK.$$

In particular $\int_0^{K_{\min}} P(K) K^{-2} dK \le P(K_{\min})/(\gamma K_{\min})$. For $w(K) = 2(1 + \ln(F/K))K^{-2}$,

$$\int_0^{K_{\min}} w(K) P(K)\,dK \le \frac{2P(K_{\min})}{K_{\min}} \Big[\frac{1 + \ln(F/K_{\min})}{\gamma} + \frac{1}{\gamma^2} \Big].$$

For the upper tail and any measurable w on $[K_{\max}, \infty)$,

$$\int_{K_{\max}}^\infty |w(K)|\, C(K)\,dK \le C(K_{\max}) K_{\max}^{\eta-1} \int_{K_{\max}}^\infty |w(K)|\, K^{1-\eta} dK;$$

in particular $\int_{K_{\max}}^\infty C(K) K^{-2} dK \le C(K_{\max})/(\eta K_{\max})$, and for $w(K) = 2(1 - \ln(K/F))K^{-2}$ with $F < K_{\max} \le eF$ and $\ell = \ln(K_{\max}/F)$,

$$\int_{K_{\max}}^\infty |w(K)|\, C(K)\,dK \le \frac{2C(K_{\max})}{K_{\max}} \Big[\frac{1 - \ell}{\eta} - \frac{1}{\eta^2} + \frac{2e^{-\eta(1-\ell)}}{\eta^2} \Big].$$

Proof. The first inequality is the lemma multiplied by $w \ge 0$ and integrated. For $w(K) = K^{-2}$, $\int_0^{K_{\min}} K^{\gamma-1} dK = K_{\min}^\gamma/\gamma$. For the logarithmic weight, integration by parts gives $\int_0^{K_{\min}} K^{\gamma-1} \ln(F/K)\,dK = K_{\min}^\gamma \ln(F/K_{\min})/\gamma + K_{\min}^\gamma/\gamma^2$, the boundary term at zero vanishing because $K^\gamma \ln(F/K) \to 0$. The upper-tail bounds follow from part (b) of the lemma. For $w(K) = K^{-2}$, $\int_{K_{\max}}^\infty K^{-1-\eta} dK = K_{\max}^{-\eta}/\eta$. For the logarithmic weight, the substitution $u = \ln(K/F)$ gives $F^{-\eta} \int_\ell^\infty |1 - u| e^{-\eta u} du$, which is split at $u = 1$ and integrated by parts. □

Remark 9. The bounds use only observed wing prices and an assumed tail exponent. A tail term whose weight has constant sign lies between 0 and its bound, taken with the sign of the weight. A sign-changing weight is split into its positive and negative parts, $w = w_+ - w_-$ with $w_\pm = \max(\pm w, 0)$; by the proposition the tail integrals against w_+ and w_- lie in $[0, B_+]$ and $[0, B_-]$, where $B_\pm$ is the bound for $w_\pm$, so the term lies in $[-B_-, B_+]$. The interval $[-B, B]$ with $B = B_+ + B_-$, the bound for $|w|$, is also valid but coarser. Because γ and η are not observed, risk-neutral variance and skewness are reported as intervals over a stated range of exponents, obtained by propagating the intervals of the individual contracts, never as point values. Between $K_{\min}$ and $K_{\max}$, P and C are taken from the calibrated smile and the integrals are computed by composite Simpson's rule, with F a node and each side integrated separately, because the out-of-the-money integrand has a kink at F where $P' - C' = 1$. With node spacing h on a side $[a, b]$ and an even number of subintervals for Simpson's rule, the composite trapezium error is at most $(b-a)h^2 \max|g''|/12$ if $g \in C^2[a,b]$, and the composite Simpson error at most $(b-a)h^4 \max|g^{(4)}|/180$ if $g \in C^4[a,b]$ (Süli and Mayers, 2003).

R5(a). Strike from pips delta at a flat volatility

This result is standard. The pips spot and forward deltas are those defined by Reiswich and Wystup (2012; working-paper version of 20 March 2010, Sections 1.3 and 1.5), and they give the closed-form strike for pips spot and forward deltas (eqs. (15) and (16) of that version); the proof is written out for completeness, and none of it is my own. It is the closed form used for pips deltas in `src/qef/fx/gk.py` and for the strikes $K^{ms}_{c,p}$ in the proof of R8.

Proposition 10. *Let $\sigma > 0$, let $D > 0$ be the base-currency discount factor (with forward delta, $D = 1$), and let $\phi = 1$ for a call and $\phi = -1$ for a put. The pips delta at the flat volatility σ, $\Delta(K) = \phi D\,\Phi\big(\phi\, d_+(K)\big)$, with $s = \sigma\sqrt{\tau}$ in d_+, is continuous and strictly decreasing in K, and maps $(0,\infty)$ onto $(0, D)$ for a call and onto $(-D, 0)$ for a put. For each Δ with the sign of ϕ and $0 < |\Delta| < D$, the unique strike with that delta is*

$$K = F \exp\Big(\tfrac{1}{2}\sigma^2\tau - \phi\,\sigma\sqrt{\tau}\,\Phi^{-1}\big(|\Delta|/D\big)\Big).$$

Proof. Since Φ takes values in $(0,1)$, $\Delta(K)$ has the sign of ϕ and $|\Delta(K)| = D\,\Phi(\phi\, d_+(K)) \in (0, D)$. The map $K \mapsto d_+(K)$ is a strictly decreasing bijection of $(0,\infty)$ onto $\mathbb{R}$, and $\Delta'(K) = \phi^2 D\,\varphi(\phi\, d_+)\, d_+'(K) = D\,\varphi(d_+)\, d_+'(K) < 0$, since φ is even. As $K \downarrow 0$, $d_+ \to \infty$, so $\Delta \to D$ for a call and $\Delta \to 0$ for a put; as $K \to \infty$, $d_+ \to -\infty$, so $\Delta \to 0$ for a call and $\Delta \to -D$ for a put. Continuity and strict monotonicity give the stated ranges and uniqueness. Given Δ with the sign of ϕ and $0 < |\Delta| < D$, the equation $\Phi(\phi\, d_+) = |\Delta|/D$ gives $\phi\, d_+ = \Phi^{-1}(|\Delta|/D)$, hence $d_+ = \phi\,\Phi^{-1}(|\Delta|/D)$ because $\phi^2 = 1$. Substituting into $\ln(F/K) = s\, d_+ - s^2/2$ gives $\ln(K/F) = s^2/2 - \phi\, s\,\Phi^{-1}(|\Delta|/D)$, which is the stated formula. □

R5(b). Premium-adjusted call delta

The premium-adjusted delta, its non-monotonicity in the strike and the market practice of taking the strike to the right of the delta maximum are described by Reiswich and Wystup (2012; working-paper version of 20 March 2010, Sections 1.3 and 1.5), who also show that the premium-adjusted delta lies below the simple delta because the call price is positive, and state that the maximum solves $\sigma\sqrt{\tau}\,\Phi(d_-) = \varphi(d_-)$. Jäckel (2020, Section 2, eq. (20)) shows that the premium-adjusted forward delta of a call has a single maximum, located where the inverse Mills ratio equals the total volatility, and takes the larger of the two solutions, which is the strike on the conventional branch. The proposition below states this shape in the notation used here; its proof is the same inverse Mills ratio argument, written out for completeness, and none of it is my own.

Let $D > 0$ and $h(K) = D\,(K/F)\,\Phi\big(d_-(K)\big)$ for $K > 0$. With spot delta, D is the base-currency discount factor; with forward delta, $D = 1$.

Lemma 11. *The inverse Mills ratio $\lambda(d) = \varphi(d)/\Phi(d)$ is strictly decreasing on $\mathbb{R}$, with $\lambda(d) \to \infty$ as $d \to -\infty$ and $\lambda(d) \to 0$ as $d \to \infty$.*

Proof. Differentiation gives $\lambda'(d) = -\lambda(d)\,[d + \lambda(d)]$, so it suffices that $d\,\Phi(d) + \varphi(d) > 0$. This function has derivative $\Phi(d) > 0$ and tends to 0 as $d \to -\infty$; hence $d\,\Phi(d) + \varphi(d) = \int_{-\infty}^{d} \Phi(u)\,du > 0$. The limits follow from $\Phi \to 0$, $\varphi \to 0$ and the Mills-ratio asymptotics $\Phi(d) \sim \varphi(d)/|d|$ as $d \to -\infty$. The lemma is known: the normal density is log-concave, so Φ is log-concave (Bagnoli and Bergstrom, 2005, Section 2.1, Theorem 1) and $\lambda = (\log \Phi)'$ is decreasing, and inequalities of this type for the Mills ratio are classical (Sampford, 1953). The short direct proof above is written out for completeness. □

Proposition 12. *The function h is continuous on $(0, \infty)$, tends to 0 as $K \downarrow 0$ and as $K \to \infty$, and satisfies $h < D$. There is a unique K^* such that h is strictly increasing on $(0, K^*]$ and strictly decreasing on $[K^*, \infty)$; $K^* = F \exp(-d^* s - s^2/2)$, where d^* is the unique solution of $\lambda(d) = s$. Each $\Delta \in (0, h(K^*))$ has exactly two preimages, one on each side of K^*.*

Proof. As $K \downarrow 0$, $d_- \to \infty$, so $h \to 0$. As $K \to \infty$, $d_- \to -\infty$ and, by the Mills-ratio asymptotics and the identity above, $K\Phi(d_-) \sim K\varphi(d_-)/|d_-| = F\varphi(d_+)/|d_-| \to 0$. Moreover $F\Phi(d_+) - K\Phi(d_-)$ is the undiscounted call price, which is positive, so $h < D\,\Phi(d_+) < D$.

Since $d_-'(K) = -1/(Ks)$,

$$h'(K) = \frac{D}{F}\Big[\Phi(d_-) - \frac{\varphi(d_-)}{s}\Big] = \frac{D\,\Phi(d_-)}{F\,s}\,\big[s - \lambda(d_-)\big].$$

By the lemma, $s - \lambda(d)$ is strictly increasing in d and changes sign exactly once, at d^*. Because d_- is strictly decreasing in K, $h' > 0$ for $K < K^*$ and $h' < 0$ for $K > K^*$, where $d_-(K^*) = d^*$. The stated formula for K^* inverts d_-. The final claim follows from the intermediate value theorem and strict monotonicity on each side. □

Remark 13. The market convention selects the preimage in $[K^*, \infty)$. At one month and 25Δ this root lies far to the right of K^*; bracketing matters at long maturities and high volatilities.

R5(c). Delta-to-strike inversion with a smile

Reiswich (2010, Section 3.8 and Fig. 3.6) shows that the delta implied by a smile need not be monotone in the strike, so that a quoted delta can correspond to several strikes or to none, and Healy (2025, Section 5.3) gives SABR smiles with $\beta = 1$ whose pips call delta is not monotone and never reaches 25Δ. Fukasawa (2012, Lemma 2.1 and Propositions 2.1–2.2) shows, by the same derivative calculation as below, that d_+ and d_- are strictly decreasing in the strike for every arbitrage-free smile, and Mingone (2023, Lemma 4.1) shows that each delta has a unique strike if and only if d_+ is strictly decreasing along the smile and maps onto the real line. The condition in the proposition is the two-sided form of Fukasawa's inequality (his Lemma 2.1), which in the notation here reads $1 + K\sqrt{\tau}\,d_-(K, \sigma(K))\,\sigma'(K) > 0$ and holds for every arbitrage-free smile; the proof of his Proposition 2.1 is the derivative identity below. Here the inequality is a hypothesis checked numerically on each fitted smile, which is not assumed to be free of arbitrage, much as Martini and Mingone (2022, Sections 3.1, 4.1 and 5.5.1) impose the same conditions on SVI slices. The proposition is stated for pips spot and forward deltas with the discount factor D; nothing in it is mine beyond this check.

Proposition 14. *Let $\sigma : (0,\infty) \to (0,\infty)$ be continuously differentiable and let $\Delta_\phi(K) = \phi D\,\Phi\big(\phi\, d_+(K,\sigma(K))\big)$ be the pips spot delta (D the base-currency discount factor) or forward delta ($D = 1$) of a call ($\phi = 1$) or put ($\phi = -1$). On an interval J, if*

$$|\sigma'(K)|\,K\sqrt{\tau}\,|d_-(K,\sigma(K))| < 1 \qquad \text{for all } K \in J,$$

then Δ_ϕ is strictly decreasing on J, and each delta attained on J is attained at exactly one strike in J.

Proof. With $d_+ = \ln(F/K)/(\sigma\sqrt{\tau}) + \sigma\sqrt{\tau}/2$, one has $\partial_K d_+ = -1/(K\sigma\sqrt{\tau})$ and $\partial_\sigma d_+ = -d_-/\sigma$. By the chain rule,

$$\frac{d}{dK}\, d_+\big(K,\sigma(K)\big) = -\frac{1}{K\sigma\sqrt{\tau}}\Big[1 + K\sqrt{\tau}\, d_-\, \sigma'(K)\Big],$$

which is negative on J under the stated condition. Since $\Delta'_\phi(K) = \phi^2 D\,\varphi(d_+)\,\frac{d}{dK}d_+ = D\,\varphi(d_+)\,\frac{d}{dK}d_+$, the function Δ_ϕ is strictly decreasing on J, and injectivity gives uniqueness. □

Remark 15. In log-moneyness $k = \ln(K/F)$, $K\sigma'(K) = \partial\sigma/\partial k$, so the condition reads $|\partial\sigma/\partial k|\sqrt{\tau}\,|d_-| < 1$. Across the calibrated one-month smiles of the primary sample, its maximum over ±4 ATM standard deviations is 0.76 (`scripts/check_reported_conditions.py`), so the smile delta-to-strike map is injective on that range at every month-end. For premium-adjusted delta $\phi D(K/F)\Phi(\phi d_-)$ the analogous derivative is $\phi(D/F)\big[\Phi(\phi d_-) + \phi K\varphi(d_-)\,\frac{d}{dK}d_-\big]$ with $\frac{d}{dK}d_- = -\big[1 + K\sqrt{\tau}\,d_+\sigma'(K)\big]/(K\sigma\sqrt{\tau})$; numerically, the premium-adjusted put delta is strictly monotone and the call delta strictly decreasing to the right of its maximum on the same range at every month-end.

R6. Diffusive null of the hedge cost

The comparison of hedged and unhedged carry returns follows Jurek (2014). For a position paired with a single put whose delta is not hedged, the construction of Burnside et al. (2011), he states that it earns a fraction $1 + \Delta_P$ of the diffusive premium, with $\Delta_P < 0$ the put's delta (working-paper version of August 2013, Section 2.2); his own hedged trades also hedge the delta of the option overlay, and he reads the resulting ratio as an upper bound on the crash share. Farhi et al. (2015, Section 5.1, eqs. (8)–(9)) obtain the same leading-order form as a small-maturity limit for their hedge. The mean-value form and the error bound below are my own short derivation.

Let the undiscounted put value with lognormal forward G and volatility σ be

$$p(G) = K\,\Phi\big(-d_-(G)\big) - G\,\Phi\big(-d_+(G)\big), \qquad d_\pm(G) = \frac{\ln(G/K) \pm s^2/2}{s}.$$

Using $K\varphi(d_-) = G\varphi(d_+)$, $p'(G) = -\Phi(-d_+(G))$.

Suppose that under $\mathbb{Q}$, S_τ is lognormal with mean F and log-variance s^2, and that under $\mathbb{P}$ it is lognormal with the same log-variance and mean $Fe^{\lambda\tau}$ (no jumps, no volatility premium). A unit long forward costs nothing and has $\mathbb{P}$-expected payoff $\mu_U = F(e^{\lambda\tau} - 1)$. Adding a put bought at its $\mathbb{Q}$-value, with the premium financed to τ, gives $\mu_H = \mu_U + p(Fe^{\lambda\tau}) - p(F)$.

Proposition 16. *If $\lambda \neq 0$, the hedge-cost ratio $\theta_0 = 1 - \mu_H/\mu_U$ satisfies $\theta_0 = \Phi(-d_+(\xi))$ for some ξ between F and $Fe^{\lambda\tau}$, and*

$$\big|\theta_0 - \Phi(-d_1)\big| \le \varphi(0)\,\frac{|\lambda|\sqrt{\tau}}{\sigma}, \qquad d_1 = d_+(F).$$

Proof. By the mean value theorem, $p(Fe^{\lambda\tau}) - p(F) = -\Phi(-d_+(\xi))\, F(e^{\lambda\tau} - 1)$ for some ξ between the two forwards, so $\theta_0 = -[p(Fe^{\lambda\tau}) - p(F)]/\mu_U = \Phi(-d_+(\xi))$. Since $|d_+(\xi) - d_+(F)| = |\ln(\xi/F)|/s \le |\lambda|\tau/s$ and Φ is Lipschitz with constant $\varphi(0)$, the bound follows. □

Remark 17. With no crash premium at all, hedging with a put of forward delta $-\Phi(-d_1)$ surrenders approximately that fraction of the carry premium: about 0.25 for a 25Δ put and 0.10 for a 10Δ put. This is the null against which the realised ratio θ_{UB} and the skew component of E3 are read.

R7. Entropy decomposition of currency premia

The decomposition is due to Backus, Foresi and Telmer (2001): with the exchange rate equal to the ratio of the two pricing kernels (their Proposition 1), the expected excess return on a forward position is the difference between the log of the expectation and the expectation of the log of the two kernels (their eq. (11)), and its higher cumulants enter only through that difference (eqs. (12)–(13)). Conditional entropy, as this difference is now called, and the entropy bound are as in Backus, Chernov and Zin (2014), who relate the bound to Bansal and Lehmann (1997) and Alvarez and Jermann (2005). Lustig and Verdelhan (2019, Section 2.5) restate the decomposition with a wedge between the exchange rate and the ratio of kernels, and Jiang, Krishnamurthy and Lustig (2021) link the value of the dollar to a convenience yield inferred from the Treasury basis, a covered-parity deviation measured with government bond yields. The identity below writes the same decomposition with the covered-interest-parity basis as the wedge on the forward side; the derivation is elementary.

Fix t and write $\mathbb{E}_t$ for $\mathbb{E}[\cdot \mid \mathcal{F}_t]$. For a positive random variable M with $\mathbb{E}_t|\log M| < \infty$, the conditional entropy is $L_t(M) = \log \mathbb{E}_t M - \mathbb{E}_t \log M \ge 0$. Let $M, M^* > 0$ be $\mathcal{F}_{t+1}$-measurable discount factors in L^2 that price domestic and foreign payoffs in their own currencies, with $\mathbb{E}_t M = e^{-r_t}$, $\mathbb{E}_t M^* = e^{-r_t^*}$, $\mathbb{E}_t|\log M|, \mathbb{E}_t|\log M^*| < \infty$ and $MS_{t+1}/S_t \in L^2$, where S is the domestic price of foreign currency. Write $m = \log M$, $\Delta s = \log(S_{t+1}/S_t)$, and $x_t = (f_t - s_t) - (r_t - r_t^*)$ for the covered-interest-parity basis. With exchange rates quoted in foreign currency per USD, as in Du, Tepper and Verdelhan (2018; NBER Working Paper 23170, eq. (4)), x_t is minus their basis.

Lemma 18. *Suppose every foreign payoff $X^* \in L^2(\mathcal{F}_{t+1})$ is traded and priced consistently in both currencies, $S_t \mathbb{E}_t[M^* X^*] = \mathbb{E}_t[M S_{t+1} X^*]$. Then $M^* = MS_{t+1}/S_t$ almost surely.*

Proof. The difference $D = M^* - MS_{t+1}/S_t$ lies in L^2 and satisfies $\mathbb{E}_t[DX^*] = 0$ for all $X^* \in L^2$. Taking $X^* = D$ gives $\mathbb{E}_t D^2 = 0$. □

Proposition 19. *Under the lemma's conclusion, (i) $\mathbb{E}_t[\Delta s + r_t^* - r_t] = L_t(M) - L_t(M^*)$ and $\mathbb{E}_t[s_{t+1} - f_t] = L_t(M) - L_t(M^*) - x_t$. (ii) If $k(u) = \log \mathbb{E}_t e^{um}$ is finite near 0 and its Maclaurin series has radius of convergence greater than 1, then $L_t(M) = \sum_{j \ge 2} \kappa_{j,t}/j!$, where $\kappa_{j,t}$ are the conditional cumulants of m. (iii) For any gross return $R > 0$ with $\mathbb{E}_t[MR] = 1$ and $\mathbb{E}_t|\log R| < \infty$, $L_t(M) \ge \mathbb{E}_t \log R - r_t$.*

Proof. (i) By definition, $\mathbb{E}_t m = -r_t - L_t(M)$ and $\mathbb{E}_t m^* = -r_t^* - L_t(M^*)$. The lemma gives $\Delta s = m^* - m$, so $\mathbb{E}_t \Delta s = r_t - r_t^* + L_t(M) - L_t(M^*)$. The second identity follows from $s_{t+1} - f_t = \Delta s - (f_t - s_t) = \Delta s - (r_t - r_t^*) - x_t$. (ii) $L_t(M) = k(1) - k'(0)$ with $k'(0) = \kappa_{1,t}$, and $k(1) = \sum_{j \ge 1} \kappa_{j,t}/j!$ under the stated convergence. (iii) By Jensen's inequality, $0 = \log \mathbb{E}_t[MR] \ge \mathbb{E}_t m + \mathbb{E}_t \log R$, so $L_t(M) = -r_t - \mathbb{E}_t m \ge \mathbb{E}_t \log R - r_t$. □

Remark 20. The identity in (i) needs only that some pair of discount factors prices bonds and exchange rates consistently; completeness makes the pair unique. Brandt, Cochrane and Santa-Clara (2006) show that the smoothness of exchange rates relative to the volatility of marginal utility implied by equity premia requires a very high degree of international risk sharing.

Options on S reveal the risk-neutral law of $\Delta s = m^* - m$, not the separate cumulants of m and m^*, so, without a parametric model of the discount factors, they constrain but do not identify the split of $L_t(M) - L_t(M^*)$ across cumulant orders; Jurek and Xu (2014), according to their abstract, calibrate such a model to G10 option prices. Hedged and unhedged carry returns give two lower bounds on the same $L_t(M)$ through (iii).

R8. Local well-posedness of the smile calibration

Reiswich (2010, Theorems 2 and 3 and Appendices A and B) uses the implicit function theorem to show that a volatility-strike function exists near the at-the-money point and that the market-strangle calibration of the simplified parabolic smile is locally solvable for the smile strangle near a zero risk reversal. The proposition below applies the same argument to the three SABR parameters and the three quotes. Le Floc'h and Kennedy (2014, Section 4, eqs. (30)–(31)) invert the leading-order expansion of the Hagan formula (Hagan et al., 2002, eq. (3.1a)) in closed form, recovering (α, ρ, ν) from the at-the-money level, skew and curvature, which is an approximate, global counterpart of the local and exact statement here. The three calibration equations are the conditions of Reiswich and Wystup (2012; working-paper version, section on the construction of implied volatility smiles, eqs. (17), (18) and (25)), to which Reiswich (2010, Section 3.2) calibrates SABR by least squares and Clark (2011, Sections 3.7 and 3.10) with $\beta = 1$ by numerical minimisation, neither with a result on existence or uniqueness, and Henrard (2011) uses the implicit function theorem in the same way for any calibration with as many instruments as parameters. In the lemma, the value of $z/x(z)$ at $z = 0$ is that of the at-the-money formula of Hagan et al. (2002, eq. (2.18)), and Choi and Wu (2021, eq. (5)) give its Taylor expansion there (their footnote 2 relates their x to the one here), and the integral representation is Hadamard's lemma (Milnor, 1963, Section 2, Lemma 2.1). What I claim as my own is the explicit statement that the $\beta = 1$ formula is jointly smooth on the stated domain, including $\nu = 0$, and the local well-posedness statement for SABR under conditions (i) to (iv), written for FX delta conventions; I have not found these in this form in the sources I consulted.

Lemma 21. *For $\beta = 1$ the Hagan, Kumar, Lesniewski and Woodward (2002) implied volatility $\sigma(K;\theta)$, $\theta = (\alpha, \rho, \nu)$, is infinitely differentiable on $\{K > 0\} \times (0, \infty) \times (-1, 1) \times [0, \infty)$.*

Proof. For $\beta = 1$, $\sigma = \alpha \big(z/x(z)\big)\big[1 + \big(\rho\nu\alpha/4 + (2 - 3\rho^2)\nu^2/24\big)\tau\big]$ with $z = (\nu/\alpha)\ln(F/K)$ and $x(z) = \ln\big[(\sqrt{1 - 2\rho z + z^2} + z - \rho)/(1 - \rho)\big]$. Since $(1 - 2\rho z + z^2) - (z - \rho)^2 = 1 - \rho^2 > 0$, the square root exceeds $|z - \rho|$, so the argument of the logarithm is positive and x is smooth. Moreover $x'(z) = (1 - 2\rho z + z^2)^{-1/2} > 0$ and $x(0) = 0$, so $x(z, \rho) = z\, h(z, \rho)$ with $h(z, \rho) = \int_0^1 (1 - 2\rho t z + t^2 z^2)^{-1/2}\, dt$. Since $1 - 2\rho t z + t^2 z^2 \geq 1 - \rho^2 > 0$, the integrand is smooth and positive on $\mathbb{R} \times (-1, 1) \times [0, 1]$, so by differentiation under the integral h is smooth and positive, and $z/x(z) = 1/h(z, \rho)$ (equal to 1 at $z = 0$) is jointly smooth in (z, ρ). Compositions and products of smooth maps are smooth. □

Fix F, τ, D and the delta convention, and write $q = (\sigma_{ATM}, RR, BF)$ for the quotes. For parameters θ, let $K_A(\theta)$ solve $\ln(K/F) \mp \sigma(K;\theta)^2\tau/2 = 0$ (the delta-neutral-straddle strike, with the sign given by the premium convention). Let $K_c(\theta)$ and $K_p(\theta)$ solve $\Delta\big(K, \sigma(K;\theta)\big) = \pm 0.25$. Let $K_c^{ms}(q)$ and $K_p^{ms}(q)$ be the $\pm 25\Delta$ strikes at the single volatility $\sigma_{ATM} + BF$. Define $\Psi = (\Psi_1, \Psi_2, \Psi_3)$ by

$$\Psi_1 = \sigma(K_A;\theta) - \sigma_{ATM}, \qquad \Psi_2 = \sigma(K_c;\theta) - \sigma(K_p;\theta) - RR,$$

$$\Psi_3 = \sum_{i \in \{c,p\}} \big[V\big(K_i^{ms}, \sigma(K_i^{ms};\theta)\big) - V\big(K_i^{ms}, \sigma_{ATM} + BF\big)\big],$$

where V is the undiscounted Garman–Kohlhagen premium of the corresponding option. The calibrated smile solves $\Psi(\theta, q) = 0$. The market-strangle condition is that of Reiswich and Wystup

(2012). For θ near θ^*, $K_A(\theta)$, $K_c(\theta)$ and $K_p(\theta)$ denote the C^1 branches through the strikes at θ^* given by the implicit function theorem under (i) and (ii) below; for a premium-adjusted call, the strike at θ^* is the root with $\partial_K \Delta < 0$ (the conventional branch), which is the root that `strike_from_delta_smile` selects.

Proposition 22. *Let $\Psi(\theta^*, q^*) = 0$. Suppose that at (θ^*, q^*): (i) $1 \mp K\sigma\sigma'(K)\tau \neq 0$ at K_A; (ii) the strike derivative of $\Delta(K, \sigma(K;\theta^*))$ is non-zero at K_c and K_p (for pips delta, for instance under the condition of R5(c); for premium-adjusted delta the derivative in the remark after R5(c) must be checked directly, since a flat smile satisfies the R5(c) condition while the call derivative vanishes at K^*); (iii) for premium-adjusted delta, the 25Δ call strike at the flat volatility $\sigma^*_{ATM} + BF^*$ lies strictly to the right of the maximiser, $K_c^{ms} > K^*$ (equivalently $0.25 < h(K^*)$ in the notation of R5(b)); and (iv) $D_\theta \Psi(\theta^*, q^*)$ is non-singular. Then there are neighbourhoods $U \ni q^*$ and $W \ni \theta^*$ and a unique C^1 map $\theta : U \to W$ with $\Psi(\theta(q), q) = 0$. In particular θ^* is the only calibration in W with strikes on these branches, and the calibrated smile varies smoothly with the quotes.*

Proof. Since x has the sign of z, $z/x(z) > 0$, so $\sigma(\cdot;\theta) > 0$ for all K exactly when $1 + \big(\rho\nu\alpha/4 + (2 - 3\rho^2)\nu^2/24\big)\tau > 0$. This holds at θ^* because $\sigma(K_A;\theta^*) = \sigma_{ATM} > 0$, hence near θ^*, and Δ and V are evaluated only at positive volatilities. By the lemma, each defining equation of K_A, K_c, K_p is then C^1 in (K, θ). Condition (i) is the non-vanishing of the K-derivative of the first; condition (ii) that of the other two. The implicit function theorem therefore makes K_A, K_c and K_p C^1 functions of θ near θ^*. The strikes K_i^{ms} depend on q through $\sigma_{ATM} + BF$: in closed form for pips delta, $K_{c,p}^{ms} = F \exp\big(\frac{1}{2}s_q^2 \mp s_q \Phi^{-1}(0.25/D)\big)$ with $s_q = (\sigma_{ATM} + BF)\sqrt{\tau}$, by R5(a); and by the implicit function theorem for premium-adjusted delta, since $h' < 0$ on (K^*, ∞) by the proof of R5(b) and condition (iii), and $\partial_K\big[-D(K/F)\Phi(-d_-)\big] = -(D/F)\big[\Phi(-d_-) + \varphi(d_-)/s\big] < 0$ for the put. Hence Ψ is C^1 in (θ, q) near (θ^*, q^*), and the implicit function theorem applied to Ψ with (iv) gives the claim. □

Remark 23. Condition (iv) is checked at every calibration through the finite-difference Jacobian J returned by the solver. J is taken in $x = (\ln\alpha, \operatorname{atanh}\rho, \ln\nu)$, with Ψ_3 divided by the flat-volatility strangle vega $\mathcal{V}$, so $J = \operatorname{diag}(1, 1, \mathcal{V}^{-1})\, D_\theta\Psi\, \operatorname{diag}(\alpha, 1-\rho^2, \nu)$, which is non-singular exactly when $D_\theta\Psi$ is. Across the primary sample $\operatorname{cond}(J)$ is at most 125 under the market-strangle reading and at most 634 under the smile-strangle reading (`scripts/check_reported_conditions.py`). Under the smile-strangle reading, Ψ_3 is replaced by $\frac{1}{2}[\sigma(K_c;\theta) + \sigma(K_p;\theta)] - \sigma_{ATM} - BF$, and the proposition and its proof hold without condition (iii). The proposition is local: it does not assert existence of a solution for arbitrary quotes, and it does not exclude other solutions far from θ^*. The implementation therefore restarts from several points and records non-convergence.

R10. Admissible tail exponents and sharp identification of moments

R4 bounds each moment contract separately, and it leaves open which exponents are compatible with the quotes. This section answers both questions under the hypotheses of R4. The notation is that of R3 and R4, with X the terminal exchange rate under the quote-currency forward measure, $P_0 = P(K_{\min})$, $G_0 = G(K_{\min}) = P'(K_{\min}+)$, $C_0 = C(K_{\max})$ and $\bar{G}_0 = \mathbb{Q}(X \geq K_{\max}) = -C'(K_{\max}-)$, all taken from the calibrated smile on $[K_{\min}, K_{\max}]$. The one-sided slopes are those inside the observed range, which the smile determines: by the difference-quotient argument in the proof of R2, $P'(K_{\min}+) = \mathbb{Q}(X \leq K_{\min})$ and $C'(K_{\max}-) = -\mathbb{Q}(X \geq K_{\max})$. An atom of X at either boundary strike is therefore not excluded by the observations, and $\bar{G}_0 = \bar{G}(K_{\max})$ when X has no atom at $K_{\max}$. Write $H_L(\gamma)$ for the hypothesis that $G(u)/u^\gamma$ is nondecreasing on $(0, K_{\min}]$ and $H_U(\eta)$, $\eta > 1$, for the hypothesis that $u^\eta \bar{G}(u)$ is nonincreasing on $[K_{\max}, \infty)$.

Several ingredients are known, and I credit them where they are used. $H_L(\gamma)$ is the restriction to the lower tail of the star order (Lando, Arab and Oliveira, 2023, Definition 2) of X against the power-function law, and $H_U(\eta)$ is the corresponding statement against the Pareto law; for these reference laws the star order coincides with the reversed hazard rate order and the hazard rate order respectively, by a direct check, since $G(u)/u^\gamma$ and $u^\eta \bar{G}(u)$ are the ratios of the distribution and survival functions to those of the reference laws. With a density, $H_L(\gamma)$ says that the elasticity $ug(u)/G(u)$ of the distribution function, studied by Veres-Ferrer and Pavía (2022, Sections 2–5), is at least γ in the tail. The quantity $K_{\min}P'(K_{\min})/P(K_{\min})$ is proposed as the natural exponent of a power-law put extrapolation by Benaim, Dodgson and Kainth (2008, footnote 6 of the version consulted), and Qin, Yang, Che and Feng (2026, Propositions 1 and 2) construct the pure power-law tails that match the boundary price and slope, whose exponents are exactly $\bar{\gamma}$ and $\bar{\eta}$ and are unique within the power-law family; their Section 5.5 also gives a necessary and sufficient condition (eq. (30)) on the boundary call price, its strike slope and the density for a shifted power-law right tail that matches all three, with a unique solution when it holds, and the check of setting (i) in (d) below is a different condition of the same kind for the classes H_L and H_U. Figlewski (2010, Section 5) completes a recovered risk-neutral density with generalised extreme value tails attached inside the quoted range, matching the tail probability at one quantile and the density there and at a second quantile, and Bollinger, Melick and Thomas (2023, Sections 1–3) require such tails also to price the options at the outermost strikes and show with scaled Weibull tails that this is feasible; (d) asks the corresponding question for H_L and H_U. Taleb, Yarckin, Mann, Delic and Spitznagel (2023, Section IV) derive from convexity at the strike where a Pareto tail is attached a restriction on its index, in their parametrisation a lower bound (their eq. (10)), which is related to, but not the same as, the admissibility condition in (a) below; Bollerslev, Todorov and Xu (2015, Section 3, eq. (3.3)) use an estimator of tail decay based on log-log slopes of out-of-the-money option prices. Sharp bounds on an option price given the prices of other options, by convex optimisation and without a tail hypothesis, go back to Bertsimas and Popescu (2002). Lam and Mottet (2017, Section 4) and Mottet and Lam (2017) obtain sharp bounds over tails with convex or higher-order monotone densities whose boundary values are fixed by the observed part, by a change of variable that writes the tail through a probability measure; that device is the one used below for the classes H_L and H_U, and their feasibility condition with a unique tail at equality (Lam and Mottet, 2017, Lemma 4.11) is the counterpart of (a) for their class. The reduction of bounds over a class generated by a family of distributions to a moment problem over the mixing measure follows Popescu (2005, Section 3); that extreme points of the resulting set are mixtures of at most two generators is Theorem 2.1 of Winkler (1988), as quoted by Pinelis (2016, Theorem 12 of the arXiv version of 2012); and the description of the bounds as sections of the convex hull of a curve is that of Kemperman (1968, Section 0). The breakdown point in (e) below is the identification breakdown point of Horowitz and Manski (1995), generalised by Stoye (2005, 2010), as set out by Masten and Poirier (2020, p. 42), and it is defined, as in Kline and Santos (2013, Example 2.3), as the level of an assumption at which the sign of a parameter stops being identified. Zhuang (2026) studies sharp bounds on option-implied tail quantities from finitely many quotes without a tail-shape hypothesis, and shows that such quotes only bound the risk-neutral probability $\mathbb{Q}(X \le K)$ at a threshold; since G_0 is that probability at $K_{\min}$, the boundary slope is itself only set-identified, an inference of mine, and here it is taken from the calibrated smile. Bondarenko, Dillschneider, Schneider and Trojani (2026), according to their abstract, derive model-free bounds on the valuation uncertainty that an observed option cross-section leaves about a risk-neutral moment under market incompleteness, and find it unbounded for essentially all standard variance, skewness and kurtosis measures.

What I claim as my own is the following. In (a), the observation that convexity at the strike where a Pareto tail is attached restricts its index is that of Taleb et al. (2023), and it is not mine; the exponents $\bar{\gamma}$ and $\bar{\eta}$ are those of the matching power-law tails of Benaim et al. (2008) and Qin

et al. (2026); what I claim is the characterisation of the exponents compatible with the observed boundary price and slope under H_L and H_U, by a condition that is necessary and sufficient within the whole class, so that $\bar\gamma$ and $\bar\eta$ are the largest admissible exponents in the whole class, together with the uniqueness of the tail within the whole class at the largest exponent. I also claim the generating family of H_L and H_U with the boundary price and slope fixed, a device that Lam and Mottet (2017) and Mottet and Lam (2017) use for other shape classes, the resulting description of the identified set of the moment contracts, and its consequences for R4; the check of setting (i) in (d); and the use of the breakdown point for the sign of option-implied skewness. I have not found these in this form in the sources I consulted.

Lemma 24. *Let $\gamma > 0$. A nondecreasing right-continuous function G on $(0, K_{\min}]$ with $G(K_{\min}) = G_0$ satisfies $H_L(\gamma)$ if and only if there is a probability measure μ on $[0,1]$ with*

$$G(K_{\min} z) = G_0\, z^{\gamma}\, \mu([0,z]), \qquad 0 < z < 1.$$

Then, for measurable ϕ with $\int_0^{K_{\min}} G|\phi|\, du < \infty$,

$$\int_0^{K_{\min}} G(u)\phi(u)\, du = G_0 K_{\min} \int_{[0,1]} \int_{z_c}^{1} z^{\gamma}\phi(K_{\min} z)\, dz\, \mu(dz_c).$$

Proof. Under $H_L(\gamma)$, $\bar h(z) = G(K_{\min} z)/(G_0 z^{\gamma})$ is nondecreasing and right-continuous on $(0,1]$ with $0 \le \bar h \le \bar h(1) = 1$. Let μ be the measure with $\mu([0,z]) = \bar h(z)$ for $0 \le z < 1$, with $\mu(\{0\}) = \bar h(0+)$, and with the remaining mass $1 - \bar h(1-)$ at 1. Conversely, for any such μ the function defined by the display is nondecreasing, right-continuous, at most G_0 and satisfies $H_L(\gamma)$, because it is a product of the nonnegative nondecreasing functions z^{γ} and $\mu([0,z])$; it tends to 0 at 0, so the law it describes has no mass at 0. The integral identity is Tonelli's theorem applied to $\mu([0,z]) = \int \mathbf{1}\{z_c \le z\}\, \mu(dz_c)$. □

The upper tail has the same structure, with $K_{\max}$, $\bar G_0$, the power $z^{-\eta}$ and a probability measure ν on $[1,\infty]$ in place of $K_{\min}$, G_0, z^{γ} and μ. Let $\eta > 1$ and $\bar G_0 > 0$. Then $\bar G$ satisfies $H_U(\eta)$ if and only if there is a probability measure ν on $[1,\infty]$ with

$$\bar G(K_{\max} z) = \bar G_0\, z^{-\eta}\, \nu((z,\infty]), \qquad z \ge 1,$$

and then, for measurable ϕ with $\int_{K_{\max}}^{\infty} \bar G|\phi|\, du < \infty$,

$$\int_{K_{\max}}^{\infty} \bar G(u)\phi(u)\, du = \bar G_0 K_{\max} \int_{[1,\infty]} \int_{1}^{z_c} z^{-\eta}\phi(K_{\max} z)\, dz\, \nu(dz_c).$$

The proof is that of the lemma, with two steps that are not pure substitutions. First, the boundary value is the mass at or above $K_{\max}$. Under $H_U(\eta)$, $\bar h_U(z) = z^{\eta}\bar G(K_{\max} z)/\bar G_0$ is nonincreasing and right-continuous on $[1,\infty)$ with $0 \le \bar h_U \le \bar h_U(1) = \mathbb{Q}(X > K_{\max})/\bar G_0 \le 1$; ν is the measure with $\nu((z,\infty]) = \bar h_U(z)$ for $z \ge 1$, so that $\nu(\{\infty\}) = \lim_{z\to\infty} \bar h_U(z)$, and with the remaining mass $1 - \bar h_U(1) = \mathbb{Q}(X = K_{\max})/\bar G_0$ at 1. An atom of ν at 1 is thus an atom of X at $K_{\max}$, as an atom of μ at 1 is an atom of X at $K_{\min}$. Conversely, for any such ν the displayed function is nonincreasing, right-continuous and at most $\bar G_0$, z^{η} times it is nonincreasing, and it tends to 0 as $z \to \infty$; with the atom $\bar G_0 \nu(\{1\})$ placed at $K_{\max}$ it describes a law on $[K_{\max}, \infty)$ of mass $\bar G_0$ that satisfies $H_U(\eta)$. Second, Tonelli's theorem is applied to $\nu((z,\infty]) = \int \mathbf{1}\{z < z_c\}\, \nu(dz_c)$, so the inner integral runs from 1 to z_c, and the point $z_c = \infty$, which carries the pure Pareto part of the tail, contributes $\int_1^{\infty} z^{-\eta}\phi(K_{\max} z)\, dz$.

For w integrable against P on $(0, K_{\min}]$, write $W(u) = \int_u^{K_{\min}} w$, so that $\int_0^{K_{\min}} wP\, dK = \int_0^{K_{\min}} GW\, du$ by Tonelli's theorem, and define on $[0,1]$

$$a(z_c) = \frac{1 - z_c^{\gamma+1}}{\gamma+1}, \qquad b_w(z_c) = \int_{z_c}^{1} z^{\gamma} W(K_{\min} z)\, dz = \frac{K_{\min}}{\gamma+1} \int_{z_c}^{1} w(K_{\min} z)\big(z^{\gamma+1} - z_c^{\gamma+1}\big) dz.$$

For w integrable against C on $[K_{\max},\infty)$, write $W_U(u)=\int_{K_{\max}}^{u} w$, so that $\int_{K_{\max}}^{\infty} wC\,dK = \int_{K_{\max}}^{\infty} \bar{G}W_U\,du$ by Tonelli's theorem, and define on $[1,\infty]$ the upper-tail counterparts

$$a_U(z_c)=\int_1^{z_c} z^{-\eta}dz=\frac{1-z_c^{1-\eta}}{\eta-1},$$
$$b_w^U(z_c)=\int_1^{z_c} z^{-\eta}W_U(K_{\max}z)\,dz=\frac{K_{\max}}{\eta-1}\int_1^{z_c} w(K_{\max}z)\big(z^{1-\eta}-z_c^{1-\eta}\big)dz,$$

with $z_c^{1-\eta}=0$ at $z_c=\infty$. By the lemma with $\phi=1$, $P_0=G_0K_{\min}\int a\,d\mu$, and with $\phi=W$, $\int_0^{K_{\min}} wP\,dK = G_0K_{\min}\int b_w\,d\mu$. By the upper-tail representation with $\phi=1$ and $\phi=W_U$, $C_0=\bar{G}_0K_{\max}\int a_U\,d\nu$ and $\int_{K_{\max}}^{\infty} wC\,dK=\bar{G}_0K_{\max}\int b_w^U\,d\nu$. Here $\eta>1$ is used: a_U is continuous and strictly increasing on $[1,\infty]$, from $a_U(1)=0$ to its supremum $a_U(\infty)=1/(\eta-1)$, which is finite only because $\eta>1$ (for $\eta\le 1$, a_U is unbounded). By contrast, a decreases from $a(0)=1/(\gamma+1)$ to $a(1)=0$ for every $\gamma>0$. In carrying the lower-tail arguments over, the roles of $K_{\min}$, P_0, G_0, a, b_w and μ on $[0,1]$ are therefore played by $K_{\max}$, C_0, $\bar{G}_0$, a_U, b_w^U and ν on $[1,\infty]$. This is a correspondence of roles, not a substitution of symbols: the normalising constant $\gamma+1$ becomes $\eta-1$, but the power $z^{\gamma+1}$ becomes $z^{1-\eta}$, the power z^{γ} becomes $z^{-\eta}$, integrals over $[z_c,1]$ become integrals over $[1,z_c]$, and W becomes W_U, so the upper-tail formulas to use are those written out above. The end points $z_c=0$ (the pure power law below $K_{\min}$) and $z_c=1$ (an atom at $K_{\min}$) correspond to $z_c=\infty$ (the pure Pareto law above $K_{\max}$) and $z_c=1$ (an atom at $K_{\max}$).

Proposition 25. *(a)* Admissible exponents. *Some law compatible with (P_0,G_0) satisfies $H_L(\gamma)$ if and only if $\gamma\le\bar{\gamma}=K_{\min}G_0/P_0-1$, and some law compatible with $(C_0,\bar{G}_0)$ satisfies $H_U(\eta)$ if and only if $1<\eta\le\bar{\eta}=1+K_{\max}\bar{G}_0/C_0$. Here $\bar{\gamma}+1$ is the strike elasticity of the put price at $K_{\min}$ and $\bar{\eta}-1$ is minus that of the call price at $K_{\max}$. At $\gamma=\bar{\gamma}$ the lower tail is unique, $G(u)=G_0(u/K_{\min})^{\bar{\gamma}}$ on $(0,K_{\min}]$, so that $P(K)=P_0(K/K_{\min})^{\bar{\gamma}+1}$ there and every tail integral is point-identified; likewise, at $\eta=\bar{\eta}$, $\bar{G}(u)=\bar{G}_0(u/K_{\max})^{-\bar{\eta}}$ for $u\ge K_{\max}$, with no atom at $K_{\max}$, so that $C(K)=C_0(K/K_{\max})^{1-\bar{\eta}}$ there. The hypotheses are nested: $H_L(\gamma)$ implies $H_L(\gamma')$ for $\gamma'<\gamma$, and $H_U(\eta)$ implies $H_U(\eta')$ for $1<\eta'<\eta$.*

(b) Sharp identified set. *Let $0<\gamma\le\bar{\gamma}$, $1<\eta\le\bar{\eta}$ and let $w_1,\dots,w_m$ satisfy $\int_0^{K_{\min}}|w_j|K^{\gamma+1}dK<\infty$ and $\int_{K_{\max}}^{\infty}|w_j|K^{1-\eta}dK<\infty$. The set of vectors $(\int_0^{K_{\min}} w_jP\,dK)_{j\le m}$ over the laws compatible with the observations and $H_L(\gamma)$ is*

$$S_L=G_0K_{\min}\Big\{\Big(\int b_{w_j}d\mu\Big)_{j\le m}:\ \mu\ \text{a probability on } [0,1],\ \int a\,d\mu=\frac{P_0}{G_0K_{\min}}\Big\},$$

and the set of vectors $(\int_{K_{\max}}^{\infty} w_jC\,dK)_{j\le m}$ over the laws compatible with the observations and $H_U(\eta)$ is

$$S_U=\bar{G}_0K_{\max}\Big\{\Big(\int b_{w_j}^U d\nu\Big)_{j\le m}:\ \nu\ \text{a probability on } [1,\infty],\ \int a_U\,d\nu=\frac{C_0}{\bar{G}_0K_{\max}}\Big\}.$$

Each is a compact convex set whose extreme points are all attained by measures with at most two atoms. The two tails vary independently, so the identified set of the raw moments (m_1,m_2,m_3) of R3 is the Minkowski sum mid $+\,S_L+S_U$, *with* mid *the observed middle part. For $m=1$, S_L is $G_0K_{\min}$ times the interval between the lower convex and upper concave envelopes, at abscissa $P_0/(G_0K_{\min})$, of the curve $z_c\mapsto(a(z_c),b_w(z_c))$, $z_c\in[0,1]$, and S_U is $\bar{G}_0K_{\max}$ times the corresponding interval, at abscissa $C_0/(\bar{G}_0K_{\max})$, for the curve $z_c\mapsto(a_U(z_c),b_w^U(z_c))$, $z_c\in[1,\infty]$.*

(c) R4 revisited. *If $w\ge 0$, the upper end of the identified interval of $\int_0^{K_{\min}} wP\,dK$ is R4's bound $P_0K_{\min}^{-(\gamma+1)}\int_0^{K_{\min}} wK^{\gamma+1}dK$, attained by a power law below $K_{\min}$ with an atom at $K_{\min}$, and the lower end is $G_0K_{\min}b_w(z^*)$ with $a(z^*)=P_0/(G_0K_{\min})$, attained by a law whose lower tail lies in $[K_{\min}z^*,K_{\min}]$. R4's upper bound is therefore sharp. If, in addition, $w>0$ on a set of*

positive measure in $(K_{\min}z^*, K_{\min})$*, the lower end is positive, so R4's lower bound of zero is not sharp. In the upper tail, if* $w \geq 0$*, the upper end of the identified interval of* $\int_{K_{\max}}^{\infty} wC\,dK$ *is R4's bound* $C_0 K_{\max}^{\eta-1} \int_{K_{\max}}^{\infty} wK^{1-\eta}dK$*, attained by a Pareto law above* $K_{\max}$ *with an atom at* $K_{\max}$*, and the lower end is* $\bar{G}_0 K_{\max} b_w^U(z_U^*)$ *with* $a_U(z_U^*) = C_0/(\bar{G}_0 K_{\max})$*, attained by a law whose upper tail lies in* $[K_{\max}, K_{\max}z_U^*]$*. If, in addition,* $w > 0$ *on a set of positive measure in* $(K_{\max}, K_{\max}z_U^*)$*, this lower end is positive and R4's lower bound of zero is not sharp in the upper tail either.*

(d) Setting (i). *Suppose* G *has a density* g *that is continuous and positive at* $K_{\min}$*, and let* $\gamma_i = K_{\min}g(K_{\min})/G_0$*. Then* $H_L(\gamma)$ *implies* $\gamma \leq \gamma_i$*, and* $H_L(\gamma_i)$ *holds if and only if the reversed hazard rate of* $\ln X$*,* $ug(u)/G(u)$ *at* $x = \ln u$*, attains its infimum over* $(0, K_{\min}]$ *at* $K_{\min}$*, which is the case if* $\ln X$ *has a log-concave distribution function. If* $H_L(\gamma_i)$ *holds, then* $\gamma_i \leq \bar{\gamma}$*. With* $\varepsilon(K) = KP'(K)/P(K)$ *and* $\gamma_i(K)$*,* $\bar{\gamma}(K)$ *the same quantities at a general strike,* $K\varepsilon'(K) = \varepsilon(K)\big(\gamma_i(K) - \bar{\gamma}(K)\big)$*, so* $\gamma_i \leq \bar{\gamma}$ *at* $K_{\min}$ *exactly when* $\ln P$ *is locally concave in* $\ln K$ *there. In the upper tail, suppose* g *is continuous and positive at* $K_{\max}$*, and let* $\eta_i = K_{\max}g(K_{\max})/\bar{G}_0$*. Then* $H_U(\eta)$ *implies* $\eta \leq \eta_i$*, and, for* $\eta_i > 1$*,* $H_U(\eta_i)$ *holds if and only if the hazard rate of* $\ln X$*,* $ug(u)/\bar{G}(u)$ *at* $x = \ln u$*, attains its infimum over* $[K_{\max}, \infty)$ *at* $K_{\max}$*, which is the case if* $\ln X$ *has a log-concave survival function. If* $H_U(\eta_i)$ *holds, then* $\eta_i \leq \bar{\eta}$*. With* $\varepsilon_U(K) = -KC'(K)/C(K)$ *and* $\eta_i(K)$*,* $\bar{\eta}(K)$ *the same quantities at a general strike,* $K\varepsilon_U'(K) = \varepsilon_U(K)\big(\bar{\eta}(K) - \eta_i(K)\big)$*, so* $\eta_i \leq \bar{\eta}$ *at* $K_{\max}$ *exactly when* $\ln C$ *is locally concave in* $\ln K$ *there, in the same sense.*

(e) Breakdown. *For* $\theta \in (0, 1]$ *let* $I(\theta)$ *be the identified set of the variance and third central moment under* $H_L(\theta\bar{\gamma})$ *and* $H_U(1 + \theta(\bar{\eta} - 1))$*. Then* $I(\theta') \subseteq I(\theta)$ *for* $\theta < \theta'$*, and* $I(1)$ *is a single point. Say that the sign of the third central moment, and hence of the skewness, is identified at* θ *if the third central moment is positive at every point of* $I(\theta)$ *or negative at every point of* $I(\theta)$*. Because the sets are nested, the set of* θ *at which the sign is identified is empty or an interval with right end* 1*. It is empty exactly when the third central moment at the single point of* $I(1)$ *is zero; there is then no breakdown point, and the sign is identified at no fraction. Otherwise its infimum* $\theta^* \in [0, 1]$ *is the breakdown point of the sign; that* $\theta^* > 0$ *is not shown here.*

Proof. (a) By the lemma, $P_0/(G_0K_{\min}) = \int a\,d\mu$, and a is continuous and strictly decreasing from $a(0) = 1/(\gamma+1)$ to $a(1) = 0$. A feasible μ therefore exists if and only if $P_0/(G_0K_{\min}) \leq 1/(\gamma+1)$, that is $\gamma \leq \bar{\gamma}$; for such γ the mixture of δ_0 and δ_1 with weight $(\gamma+1)P_0/(G_0K_{\min})$ on δ_0 is feasible. At $\gamma = \bar{\gamma}$ the constraint reads $\int a\,d\mu = a(0)$, which forces $\mu = \delta_0$ because $a < a(0)$ on $(0, 1]$. In the upper tail, $C_0/(\bar{G}_0K_{\max}) = \int a_U\,d\nu$ by the upper-tail representation, and a_U is continuous and strictly increasing from $a_U(1) = 0$ to its supremum $a_U(\infty) = 1/(\eta-1)$, finite because $\eta > 1$. A feasible ν therefore exists if and only if $C_0/(\bar{G}_0K_{\max}) \leq 1/(\eta-1)$, that is $\eta \leq \bar{\eta}$; for such $\eta > 1$ the mixture of δ_1 and δ_∞ with weight $(\eta-1)C_0/(\bar{G}_0K_{\max})$ on δ_∞ is feasible. At $\eta = \bar{\eta}$ the constraint reads $\int a_U\,d\nu = a_U(\infty)$, which forces $\nu = \delta_\infty$ because $a_U < a_U(\infty)$ on $[1, \infty)$; the representation then gives $\bar{G}(u) = \bar{G}_0(u/K_{\max})^{-\bar{\eta}}$ for $u \geq K_{\max}$ and, integrating, $C(K) = \bar{G}_0K_{\max}(K/K_{\max})^{1-\bar{\eta}}/(\bar{\eta}-1) = C_0(K/K_{\max})^{1-\bar{\eta}}$. The lower limit $\eta > 1$ is part of $H_U(\eta)$, and $\bar{\eta} > 1$ always, since $C_0, \bar{G}_0 > 0$. The elasticity statements are the definitions, since $P'(K_{\min}+) = G_0$ and $C'(K_{\max}-) = -\bar{G}_0$. Nesting holds because $G/u^{\gamma'} = (G/u^{\gamma})\,u^{\gamma-\gamma'}$ is a product of nonnegative nondecreasing functions, and, for $1 < \eta' < \eta$, $u^{\eta'}\bar{G} = (u^{\eta}\bar{G})\,u^{\eta'-\eta}$ is a product of nonnegative nonincreasing functions.

(b) By the lemma the map $\mu \mapsto$ law is a bijection between the probability measures on $[0, 1]$ satisfying the constraint and the laws below $K_{\min}$ compatible with (P_0, G_0) and $H_L(\gamma)$; each such law extends to a law on $(0, \infty)$ compatible with the observed prices, because the law on $(K_{\min}, K_{\max})$ is fixed by the smile and the mass and the mean F are fixed by P_0, G_0, C_0 and $\bar{G}_0$ (R2). The integrability conditions make a and b_{w_j} continuous on $[0, 1]$, by dominated convergence with the bound $|w_j(K_{\min}z)|z^{\gamma+1}$, so the feasible set of μ is weak* compact and convex and its image under the continuous linear map $\mu \mapsto (\int b_{w_j}d\mu)_j$ is compact and convex. Extreme points of the image are images of extreme points of the feasible set, which are mixtures of at most two Dirac measures (Winkler, 1988, Theorem 2.1, as quoted by Pinelis, 2016; its hypotheses hold because the probability measures on a compact metric space form a Choquet simplex whose

extreme points are the Dirac measures). For $m = 1$ the set of pairs $(\int a\,d\mu, \int b_w\,d\mu)$ over all probability measures is the convex hull of the compact curve $\{(a(z_c), b_w(z_c))\}$ (Kemperman, 1968), and its section at abscissa $P_0/(G_0K_{\min})$ is the stated interval. The upper tail is the same argument on $[1,\infty]$, which is compact (the map $z \mapsto 1/z$ takes it onto $[0,1]$), with the upper-tail representation in place of the lemma. The only step that is not a pure substitution is continuity at $z_c = \infty$: a_U is continuous there because $\eta > 1$, and $b^U_{w_j}$ is continuous on $[1,\infty]$ by dominated convergence with the bound $|w_j(K_{\max}z)|z^{1-\eta}$, which is integrable on $[1,\infty)$ by the integrability condition. The lower-tail law and the upper-tail law can be chosen separately, which gives the Minkowski sum.

(c) For $w \geq 0$, W is nonnegative and nonincreasing. Along the curve, $db_w/dz_c = -z_c^{\gamma}W(K_{\min}z_c)$ and $da/dz_c = -z_c^{\gamma}$, so b_w is a function of a with slope $W(K_{\min}z_c)$, which increases as a increases. The curve is therefore the graph of a convex function of a: its lower envelope is the curve itself, attained by the single atom z^*, and its upper envelope is the chord from $(0,0)$ at $z_c = 1$ to $(a(0), b_w(0))$ at $z_c = 0$. At abscissa $P_0/(G_0K_{\min})$ the chord gives $G_0K_{\min}\cdot(\gamma+1)\frac{P_0}{G_0K_{\min}}\,b_w(0) = (\gamma+1)P_0\int_0^{K_{\min}} w(K)\frac{(K/K_{\min})^{\gamma+1}}{\gamma+1}dK$, which is R4's bound. The law is the mixture of δ_0 and δ_1 of the proof of (a). If $w > 0$ on a set of positive measure in $(K_{\min}z^*, K_{\min})$, then $b_w(z^*) > 0$, because $z^{\gamma+1} - z^{*\gamma+1} > 0$ for $z > z^*$; here $z^* < 1$ because $P_0 > 0$.

In the upper tail, for $w \geq 0$, W_U is nonnegative and nondecreasing. Along the curve, $db^U_w/dz_c = z_c^{-\eta}W_U(K_{\max}z_c)$ and $da_U/dz_c = z_c^{-\eta}$, so b^U_w is a function of a_U with slope $W_U(K_{\max}z_c)$, which increases as a_U increases, since both increase with z_c. The curve is therefore again the graph of a convex function, now on $[0, 1/(\eta-1)]$: its lower envelope is the curve itself, attained by the single atom z^*_U, which exists and is unique because a_U is a continuous strictly increasing map of $[1,\infty]$ onto $[0,1/(\eta-1)]$ and $C_0/(\bar{G}_0K_{\max})$ lies in that interval by (a); its upper envelope is the chord from $(0,0)$ at $z_c = 1$ to $(1/(\eta-1), b^U_w(\infty))$ at $z_c = \infty$. At abscissa $C_0/(\bar{G}_0K_{\max})$ the chord gives

$$\begin{aligned}\bar{G}_0K_{\max}\cdot(\eta-1)\frac{C_0}{\bar{G}_0K_{\max}}\,b^U_w(\infty) &= C_0K_{\max}\int_1^{\infty} w(K_{\max}z)\,z^{1-\eta}dz \\ &= C_0K_{\max}^{\eta-1}\int_{K_{\max}}^{\infty} w(K)K^{1-\eta}dK,\end{aligned}$$

which is R4's bound. The law is the mixture of δ_1 and δ_∞ of the proof of (a): an atom of mass $\bar{G}_0 - (\eta-1)C_0/K_{\max}$ at $K_{\max}$ and $\bar{G}(u) = (\eta-1)(C_0/K_{\max})(u/K_{\max})^{-\eta}$ for $u \geq K_{\max}$. The single atom $\nu = \delta_{z^*_U}$ gives $\bar{G}(K_{\max}z) = \bar{G}_0z^{-\eta}$ for $1 \leq z < z^*_U$ and 0 beyond, so the upper tail lies in $[K_{\max}, K_{\max}z^*_U]$. If $w > 0$ on a set of positive measure in $(K_{\max}, K_{\max}z^*_U)$, then $b^U_w(z^*_U) > 0$, because $z^{1-\eta} - z^{*1-\eta}_U > 0$ for $z < z^*_U$; here $z^*_U > 1$ because $C_0 > 0$.

(d) $\ln(G/u^{\gamma})$ is nondecreasing on $(0, K_{\min}]$ and differentiable from the left at $K_{\min}$ with derivative $g/G - \gamma/u$, which is therefore nonnegative: $\gamma \leq \gamma_i$. With a density, $\ln G$ is locally absolutely continuous where $G > 0$, so $H_L(\gamma)$ holds if and only if $ug(u)/G(u) \geq \gamma$ almost everywhere on $(0, K_{\min}]$; with $\gamma = \gamma_i$ this says that the infimum is attained at $K_{\min}$. If $\ln X$ has a log-concave distribution function, its reversed hazard rate is nonincreasing and the infimum over $x \leq \ln K_{\min}$ is attained at the end point; a continuously differentiable log-concave density suffices by Bagnoli and Bergstrom (2005, Theorem 1). The inequality $\gamma_i \leq \bar{\gamma}$ is (a) applied to γ_i. Finally, $\varepsilon = KG/P$ gives $K\varepsilon' = \varepsilon + \varepsilon Kg/G - \varepsilon^2 = \varepsilon(\gamma_i(K) + 1 - \varepsilon) = \varepsilon(\gamma_i(K) - \bar{\gamma}(K))$, and $K\varepsilon'(K)$ is the derivative of ε with respect to $\ln K$.

In the upper tail, a density continuous at $K_{\max}$ excludes an atom there, so $\bar{G}(K_{\max}) = \bar{G}_0$. $\ln(u^{\eta}\bar{G})$ is nonincreasing on $[K_{\max}, \infty)$ and differentiable from the right at $K_{\max}$ with derivative $\eta/u - g/\bar{G}$, which is therefore nonpositive: $\eta \leq \eta_i$. With a density, $\ln\bar{G}$ is locally absolutely continuous where $\bar{G} > 0$, so $H_U(\eta)$ holds if and only if $ug(u)/\bar{G}(u) \geq \eta$ almost everywhere on $[K_{\max}, \infty)$ where $\bar{G} > 0$; at $x = \ln u$ this ratio is the hazard rate of $\ln X$, and with $\eta = \eta_i$ the condition says that its infimum is attained at $K_{\max}$. If $\ln X$ has a log-concave survival function,

its hazard rate is nondecreasing and the infimum over $x \geq \ln K_{\max}$ is attained at the end point; a continuously differentiable log-concave density of $\ln X$ suffices, by Bagnoli and Bergstrom (2005, Theorem 1) applied to $-\ln X$, whose distribution function is $x \mapsto \mathbb{Q}(\ln X \geq -x)$. The inequality $\eta_i \leq \bar{\eta}$ is (a) applied to η_i. Finally, $\varepsilon_U = K\bar{G}/C$ and $C' = -\bar{G}$ give $K\varepsilon_U' = \varepsilon_U - \varepsilon_U K g/\bar{G} + \varepsilon_U^2 = \varepsilon_U(1 + \varepsilon_U - \eta_i(K)) = \varepsilon_U(\bar{\eta}(K) - \eta_i(K))$, and $-\varepsilon_U$ is the derivative of $\ln C$ with respect to $\ln K$, so $\eta_i \leq \bar{\eta}$ exactly when $d^2 \ln C/d(\ln K)^2 \leq 0$ at $K_{\max}$.

(e) By the nesting in (a), the laws compatible with the hypotheses at θ' are compatible with them at θ, so the identified sets are nested; at $\theta = 1$ both tails are unique by (a), so $I(1)$ is a single point, and every $I(\theta)$ contains it. If the sign is identified at θ, it is identified at every $\theta' > \theta$, because $I(\theta') \subseteq I(\theta)$; the set is therefore empty or an interval with right end 1. If the third central moment at the point of $I(1)$ is nonzero, the sign is identified at $\theta = 1$ and the set is not empty. If it is zero, every $I(\theta)$ contains a point at which the third central moment is zero, so the sign is identified at no θ. □

Remark 26. The implementation (`src/qef/fx/identification.py`) restricts μ and ν to a grid of step points graded towards the boundary, which gives an inner approximation that converges as the grid is refined. For a fixed mean, the ranges of the variance and of the third central moment are computed exactly for that grid by linear programmes in the measures. Their extremes over the mean are not computed exactly: `SharpMomentSet._extreme` evaluates them at 25 means spread over the range of the mean (15 in the bisection for the breakdown point of (e)) and refines around the two best by bounded scalar minimisation. Every mean it evaluates is the mean of a feasible law, and every value it returns is attained by such a law, so the computed ranges are inner approximations of the sharp set for the step grid; they can only err towards declaring the sign of the skewness identified. A check with eight times as many means on 40 currency-months is reported in `reports/stage4.md`. The weights of the moment contracts of R3 satisfy the integrability conditions of (b) for every $\gamma > 0$ and $\eta > 1$. Because $I(1)$ is a point, the Pareto tails of (a) give the unique moments at the strongest admissible hypothesis; the breakdown point, when it exists, measures how far below that hypothesis the sign of the skewness survives. In the model validation of `reports/model_validation.md`, the model's own tail exponents lie between 0.58 and 0.97 of the maximal ones, setting (i) fails in seven of eight cases, and the identified set at the model's own exponents contains the true moments in every case, as (b) requires.

An observed price beyond a boundary strike tests the hypotheses and, if consistent, adds a linear constraint on the mixing measure. The following consequence of the lemma is used for the out-of-sample test with 5Δ quotes.

Proposition 27. *Let $0 < K < K_{\min}$, $k = K/K_{\min}$ and $0 \leq \gamma \leq \bar{\gamma}$, where $\gamma = 0$ stands for the limit of the weakest hypothesis. Under $H_L(\gamma)$ the identified interval of $P(K)$ is*

$$\Big[\max\Big(0,\ P_0 - G_0 K_{\min}\,\frac{1 - k^{\gamma+1}}{\gamma + 1}\Big),\ P_0 k^{\gamma+1}\Big].$$

For $K > K_{\max}$, $k = K/K_{\max}$ and $1 \leq \eta \leq \bar{\eta}$, the identified interval of $C(K)$ under $H_U(\eta)$ is $[\max(0, C_0 - \bar{G}_0 K_{\max} a_k^U),\ C_0 k^{1-\eta}]$ with $a_k^U = (1 - k^{1-\eta})/(\eta - 1)$, read as $\ln k$ at $\eta = 1$, where $\eta = 1$ again stands for the limit of the weakest hypothesis. At the limits $\gamma = 0$ and $\eta = 1$ the upper ends are approached but not attained. The upper ends decrease and the lower ends increase with the exponent, so an observed price at K is consistent with the hypothesis exactly for exponents in an interval that starts at the weakest hypothesis and may be empty.

Proof. By the lemma with $\phi = \mathbf{1}\{u \leq K\}$, $P(K) = \int_0^{K_{\min}} G\mathbf{1}\{u \leq K\}du = G_0 K_{\min} \int b\, d\mu$ with $b(z_c) = \int_{z_c}^1 z^\gamma \mathbf{1}\{z \leq k\}dz = \max(0, a(z_c) - a_k)$, where $a_k = (1 - k^{\gamma+1})/(\gamma + 1)$. As a function of a this is convex and piecewise linear, so the lower envelope at $P_0/(G_0 K_{\min})$ is the function itself, which gives the lower end, and the upper envelope is the chord from $(0, 0)$ to $(a(0), b(0)) = (1/(\gamma + 1), k^{\gamma+1}/(\gamma + 1))$, which gives $P_0 k^{\gamma+1}$, R4's bound. Since $a_k = \int_k^1 z^\gamma dz$ decreases in γ

for $k < 1$, the lower end increases in γ, and $P_0 k^{\gamma+1}$ decreases. In the upper tail, the upper-tail representation with $\phi = \mathbf{1}\{u \geq K\}$ gives $C(K) = \int_{K_{\max}}^{\infty} \bar{G}\,\mathbf{1}\{u \geq K\}du = \bar{G}_0 K_{\max} \int b^U \, d\nu$ with $b^U(z_c) = \int_1^{z_c} z^{-\eta}\mathbf{1}\{z \geq k\}dz = \max(0, a_U(z_c) - a_k^U)$, where $a_k^U = a_U(k) = \int_1^k z^{-\eta}dz$. For $\eta > 1$ this is again convex and piecewise linear in a_U, so the lower envelope at $C_0/(\bar{G}_0 K_{\max})$ gives the lower end, and the upper envelope is the chord from $(0,0)$ at $z_c = 1$ to $(a_U(\infty), b^U(\infty)) = (1/(\eta-1), k^{1-\eta}/(\eta-1))$ at $z_c = \infty$, which gives $\bar{G}_0 K_{\max} \cdot (\eta-1)\frac{C_0}{\bar{G}_0 K_{\max}} \cdot \frac{k^{1-\eta}}{\eta-1} = C_0 k^{1-\eta}$, R4's bound. Since a_k^U decreases in η for $k > 1$, the lower end increases in η, and $C_0 k^{1-\eta}$ decreases. At $\eta = 1$, $a_U(z_c) = \ln z_c$ is unbounded and the chord has no right end point. The lower end is unchanged, with $a_k^U = \ln k$, while the upper end $C_0 = \lim_{\eta\downarrow 1} C_0 k^{1-\eta}$ is approached by mixtures of δ_1 and δ_z as $z \to \infty$ but not attained: $C(K) = C_0$ would require $\bar{G} = 0$ on $(K_{\max}, K)$, hence $C_0 = 0$.

The lower tail at $\gamma = 0$ is similar. This case lies outside the lemma's hypothesis $\gamma > 0$: $H_L(0)$ says only that G is nondecreasing, and the representation of the lemma still holds with $z^0 = 1$, except that the displayed function now tends to $G_0\mu(\{0\})$ as $z \downarrow 0$, so an atom of μ at 0 would be an atom of X at 0, which R3 excludes ($X > 0$). The feasible μ therefore have no atom at 0, while the chord to $z_c = 0$ that gives the upper end needs one. The lower end is unchanged, and it is attained when $\bar{\gamma} > 0$, by the single atom at $z^* = \bar{\gamma}/(\bar{\gamma}+1) > 0$. The upper end $P_0 k = \lim_{\gamma\downarrow 0} P_0 k^{\gamma+1}$ is approached by mixtures of δ_1 and δ_z as $z \downarrow 0$ but not attained, as C_0 is not attained at $\eta = 1$: since G is nondecreasing, $P(K) = P_0 k$, that is, equal averages of G over $(0, K)$ and $(0, K_{\min})$, would require G to be constant almost everywhere on $(0, K_{\min})$, hence equal to $G(0+) = \mathbb{Q}(X \leq 0) = 0$, and then $P_0 = 0$. □

Since $H_L(0)$ holds for every law, the interval at $\gamma = 0$, $[\max(0, P_0 - G_0(K_{\min} - K)),\ P_0 K/K_{\min}]$, is the one that convexity of P and $P(0) = 0$ give alone: the supporting line of slope G_0 at $K_{\min}$ and the chord through the origin. In the upper tail $H_U(1)$, that $u\bar{G}(u)$ is nonincreasing, is a restriction, and at $\eta = 1$ the lower end $C_0 - \bar{G}_0 K_{\max} \ln k$ exceeds the convexity bound $C_0 - \bar{G}_0(K - K_{\max})$, since $\ln k < k - 1$ for $k > 1$.

Remark 28. In the out-of-sample test (`scripts/estimate_wing_test.py`) the 5Δ put and call prices also enter the identified set as equality constraints, one on each mixing measure, with the row $\max(0, a(z_c) - a_k)$ in the lower tail and its counterpart in the upper tail, and the breakdown point is searched along the common path $\gamma = \theta\bar{\gamma}$, $\eta = 1 + \theta(\bar{\eta} - 1)$. Write $I_5(\theta)$ for the constrained identified set, and $\theta_5 = \min(\theta_5^L, \theta_5^U)$, where θ_5^L and θ_5^U are the largest fractions of the maximal exponents at which the 5Δ put and call prices lie in their intervals above. Two parts of R10 do not carry over. First, (e) is proved without extra constraints, where $I(1)$ is a single point. The constraints do not depend on θ, so the sets $I_5(\theta)$ are still nested, and they are not empty exactly for $\theta \leq \theta_5$; the fractions in $(0, \theta_5]$ at which the sign is identified are therefore empty or an interval with right end θ_5. Above θ_5 there is no compatible law, and $I_5(\theta_5)$ is in general not a single point, so the interval structure of (e), with right end 1 and its criterion for when the set is empty, is not proved in this case. Second, each mixing measure is now subject to two linear constraints besides its total mass, so by Winkler's theorem with two constraints (Winkler, 1988, Theorem 2.1, as quoted by Pinelis, 2016, Theorem 12(b) of the arXiv version) the extreme points of its feasible set are mixtures of at most three Dirac measures, and three atoms occur: if $\gamma < \bar{\gamma}$ and the 5Δ put price lies strictly inside its interval, the point $\big(P_0, P(K)\big)/(G_0 K_{\min})$ lies in the interior of the triangle with vertices $(a(z_c), \max(0, a(z_c) - a_k))$ at $z_c = 0, k, 1$, and the mixture of δ_0, δ_k and δ_1 that matches both prices is an extreme point, because the vectors $(1, a(z_c), \max(0, a(z_c) - a_k))$ at these three points are linearly independent. The tail sets remain compact and convex, but the two-atom description of (b) does not hold. With the constraints, the breakdown point θ_5^* is therefore defined computationally, as the script computes it. Let $\theta_{\max} = 0.999\,\theta_5$. If $\theta_{\max} \leq 0.01$ or the sign is not identified at $\theta_{\max}$, θ_5^* is undefined; if the sign is identified at $\theta = 0.01$, $\theta_5^* = 0.01$; otherwise θ_5^* is the smallest fraction found by bisection on $[0.01, \theta_{\max}]$, with tolerance 0.01, at which the sign is identified, each evaluation being computed as in the remark after the proof of

R10, with 25 means and the two extra constraints. The bisection relies on the nesting, which holds for the sharp sets on $(0, \theta_5]$.

The Garman–Kohlhagen PDE

The equation, and the representation of its solution as an expectation, are standard (Garman and Kohlhagen, 1983; for the Feynman–Kac argument, Shreve, 2004). The verification and the uniqueness argument below are written out for completeness.

Let $dS_t = (r_d - r_f)S_t\,dt + \sigma S_t\,dW_t$ under $\mathbb{Q}^d$ with constant coefficients, and let $c(t,S)$ be the Garman and Kohlhagen (1983) call price with strike K and expiry T,

$$c(t,S) = Se^{-r_f(T-t)}\Phi(d_+) - Ke^{-r_d(T-t)}\Phi(d_-), \qquad d_\pm = \frac{\ln(S/K) + (r_d - r_f \pm \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}.$$

Proposition 29. *(a) $c \in C^{1,2}([0,T)\times(0,\infty))$ satisfies*

$$\partial_t c + (r_d - r_f)S\,\partial_S c + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} c - r_d c = 0,$$

$0 \le c \le Se^{-r_f(T-t)}$, and $c(t,S) \to (S-K)^+$ as $t \uparrow T$, locally uniformly in S. (b) If $u \in C^{1,2}([0,T)\times(0,\infty)) \cap C([0,T]\times(0,\infty))$ solves the same equation with $u(T,S) = (S-K)^+$ and $|u(t,S)| \le a(1+S^m)$ for some a, m, then $u = c$. No boundary condition at $S = 0$ is required.

Proof. (a) Write $\tau = T - t$. The identity $Se^{-r_f\tau}\varphi(d_+) = Ke^{-r_d\tau}\varphi(d_-)$ gives $\partial_S c = e^{-r_f\tau}\Phi(d_+)$ and $\partial_{SS} c = e^{-r_f\tau}\varphi(d_+)/(S\sigma\sqrt{\tau})$. The same identity gives $\partial_t c = r_f Se^{-r_f\tau}\Phi(d_+) - r_d Ke^{-r_d\tau}\Phi(d_-) - Se^{-r_f\tau}\varphi(d_+)\sigma/(2\sqrt{\tau})$. Substitution verifies the equation. The bounds follow from $c = e^{-r_d\tau}\mathbb{E}[(S_T - K)^+ \mid S_t = S]$ and $\mathbb{E}[S_T \mid S_t = S] = Se^{(r_d - r_f)\tau}$. As $\tau \downarrow 0$, $\Phi(d_\pm) \to \mathbf{1}\{S > K\}$ for $S \ne K$ and $c(t,K) \to 0$, and c is Lipschitz in S with constant $e^{|r_f|T}$ uniformly in t, which gives locally uniform convergence.

(b) Fix (t,S) and let $\tau_n = \inf\{s \ge t : S_s \notin (1/n, n)\} \wedge (T - 1/n)$. Itô's formula for $e^{-r_d(s-t)}u(s,S_s)$ on $[t,\tau_n]$, the equation, and the boundedness of $\partial_S u$ on compact sets give $u(t,S) = \mathbb{E}\big[e^{-r_d(\tau_n - t)}u(\tau_n, S_{\tau_n})\big]$. Because S is a geometric Brownian motion started at $S > 0$, it reaches neither 0 nor ∞ before T, so $\tau_n \to T$ almost surely. Continuity of u on $[0,T]\times(0,\infty)$ then gives $u(\tau_n, S_{\tau_n}) \to (S_T - K)^+$. The growth bound and $\mathbb{E}\sup_{s\le T} S_s^m < \infty$ allow dominated convergence, so $u(t,S) = e^{-r_d(T-t)}\mathbb{E}(S_T - K)^+ = c(t,S)$. The representation uses only values in the interior $(0,\infty)$, which the process never leaves, so no condition at $S = 0$ enters. Itô's formula and the Feynman–Kac representation are standard results, stated for example in Shreve (2004). □

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